SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2. BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	1	2	1	2
Explanation	25%	1	1	1	1
Application	20%	1	1	1	1
Organization	15%	1	1	2	1
Accuracy	10%	1	1	1	1
Clarity	5%	3	1	1	1

Total Marks 29 /40

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

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REGISTER NO : 192521054

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29/30

Answer The Following Questions:

Before Transmission the presentation layer Compresses a 900MB file by 40%. The compressed data is encrypted adding 5% overhead to the compressed size. Calculate the compressed file size the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation layer improves transmission efficiency and security.

An FTP application transfer 3 GB of data over a network with an effective throughput of 120 Mbps The Application layer divides the data into requests of 3MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the application layer provides reliable services to users.

A Packet experiences the following processing delays at different OSI layers:

Cat Application Layer = 4ms, Transport layers = 3ms, Network layer = 5ms, Data Link layer = 2ms, and Physical layer = 1ms the propagation delay is 18ms and the transmission delay is 12ms. Calculate the total end to end delay experienced by the packet and explain how each layer contributes to the communication process.

A network transfer 80MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total numbers of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance

Compressed file size

Original file size : 900MB

Reduction Percentage : 40%

Calculation :

the remaining file size is 100% - 40%

100 - 40 = 60%

It is the orginal file size

Compressed size = 900MB × 0.60 = 540MB

Encrypted file size

Encryption Overhead : 5% added to the compressed data size

Calculation

Encrypted size = 540MB × (1+0.05) = 540 × 1.05

= 564MB

Total Percentage Reduction

Total Reduction in size : 900MB - 567MB = 333MB

Calculation : total Percentage Reduction = $\left(\frac{333\,MB}{900\,MB}\right)$

Contributions:

Transmission Efficiency:

Compression reduces the size of the data from 900MB to 810MB (and eventually 567MB after encryption). Because the total data payload is significantly smaller, it requires less network bandwidth and takes less time to transmit over the network

Security: Encryption scrambles the compressed data into an unreadable formate (Ciphertext) Even if an unauthorized attacker intercepts the data during transit, they cannot understand or access the original content without the proper decryption key

7. 1. Total number of Requests:

total Data : 3GB = 3x1024 MB = 3072 MB

Request size : 3MB

Calculation:

$$\text{total Requests} = \frac{3072\ MB}{3\ MB} = 1024\ req$$

2. Total transmission Time:

total data in Bits: 3GB = 3x8 = 24

Gb = 24,000MB (or 24×10^9 bits)

Network throughput : 120 Mbps = 120Mb/s

OSI Layer

Application layer: Interacts directly with software applications to initiate network service requests.

Transport layer: Handles host-to-host communication, flow control, and data segmentation / reliability

Network layer: Manages logical addressing and routing of packets across networks.

Data link layer: Handles physical node to node framing, error checking and media access control

Physical layer: Transmits raw unstructured bit streams over the Physical medium

1. Total number of Frames Required

Total useful data: 80 MB = 80×10^6 bytes

Total Frame size: 1600 bytes

Overhead per frame: 80 bytes

Payload per frame: 1600 bytes - 80 bytes = 1520 bytes

Calculation:

$$\text{total frame: } \frac{80,000,000 \text{ bytes}}{1520 \text{ bytes}} \approx 52,632 \text{ frames}$$

2. Total Overhead Transmitted

total Overhead = $52,632 \times 80$ bytes $\approx 4,210,560$ bytes

≈ 4.21 MB

Calculation: Using standard decimal networking units (1GB = 10^9 bytes):

Data = $3 \times 10^9 \times 8 = 24 \times 10^9$ bits

throughput = 120×10^6 bits / seconds

Transmission Time : $\frac{24 \times 10^9}{120 \times 10^6}$ = 200 seconds

Application layer Reliability:

The application layer Provides reliable services by utilizing reliable underlying transport protocol like TCP. It manages connection handshakes, ensure data segments are reassembled in the correct order, and handles retransmission protocols if data packets are lost or corrupted during transit

1. Total End - to End Delay

total Processing Delay : 4ms + 8ms + 5ms + 2ms + 1ms =
= 15ms

2. Propogation delay : 18 ms

transmission Delay : 12ms

3. Calculation :

total Delay = Processing Delays + Propogation delay + Transmission delay.

ToTal delay = 15ms + 18ms + 12ms = 45ms.

Transmission Efficiency

total data sent = Useful Data + Overhead

= 80MB + 1.21 MB = 81.21MB

Calculation:

$$\text{Efficiency} = \left(\frac{\text{Useful Payload}}{\text{Total frame size}}\right) \times 100\%$$

$$= \frac{1520}{1600} \times 100\% = 95\%$$

Performance Impact:

Protocol Overhead reduces overall network throughput because bandwidth is consumed by headers and trailers rather than actual data. High overhead reduces useful transmission efficiency, requiring more frames and higher network congestion to transfer the same amount of useful data.

of 3 **MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**. and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand